Assignment 4 Report

Optimizing Transformer Translation with Ray Tune & Optuna

Roll No: M25CSA031

1. Baseline Metrics

The baseline model was trained for 100 epochs with hardcoded hyperparameters on English-Hindi sentence pairs. No architectural changes were made.

Metric	Value
Training Time	42 min 29.8 sec
Final Train Loss	0.0492
BLEU Score	18.32
Epochs	100
Batch Size	60
Learning Rate	1e-4
d_model / num_heads / d_ff	512 / 8 / 2048
Dropout	0.1

2. Hyperparameter Search Space

Ray Tune was paired with OptunaSearch (TPE sampler) to intelligently explore 5 hyperparameters. ASHAScheduler pruned underperforming trials after 5 epochs (grace_period=5, reduction_factor=2, max_t=30).

Hyperparameter	API Method	Range / Choices
Learning Rate	tune.loguniform()	1e-5 to 1e-3
Batch Size	tune.choice()	16, 32, 64
Num Attention Heads	tune.choice()	4, 8 (both divide 512)
Feed-Forward Dim (d_ff)	tune.choice()	1024, 2048
Dropout Rate	tune.uniform()	0.1 to 0.4

3. Best Configuration Found

Best sweep loss: 0.1307 achieved by trial train_tune_9fcee275 in 30 epochs.

Hyperparameter	Best Value
Learning Rate	0.000279
Batch Size	64
Num Attention Heads	8
Feed-Forward Dim (d_ff)	1024
Dropout	0.101

4. Final Model Metrics & Comparison (Part 3)

The best configuration was retrained from scratch for 50 epochs. Goal: match or exceed baseline BLEU in significantly fewer epochs.

Metric	Baseline (100 ep.)	Best Model (50 ep.)	Result
Training Time	42 min 29.8 sec	~21 min (est.)	~50% faster
Final Loss	0.0492	0.0798	Comparable
BLEU Score	18.32	18.48	GOAL MET (+0.16)
Epochs	100	50	50 epochs saved

5. Key Findings

The Ray Tune + Optuna sweep consistently found that a larger batch size (64), higher learning rate (~3e-4), and smaller d_ff (1024) outperformed the baseline across the top 5 trials. Reducing d_ff from 2048 to 1024 significantly sped up training without hurting translation quality.

The best model achieved BLEU 18.48 in just 50 epochs versus the baseline's 18.32 in 100 epochs, meeting the efficiency goal. ASHA pruning eliminated approximately 8 underperforming trials early, saving roughly 40% of total sweep compute.